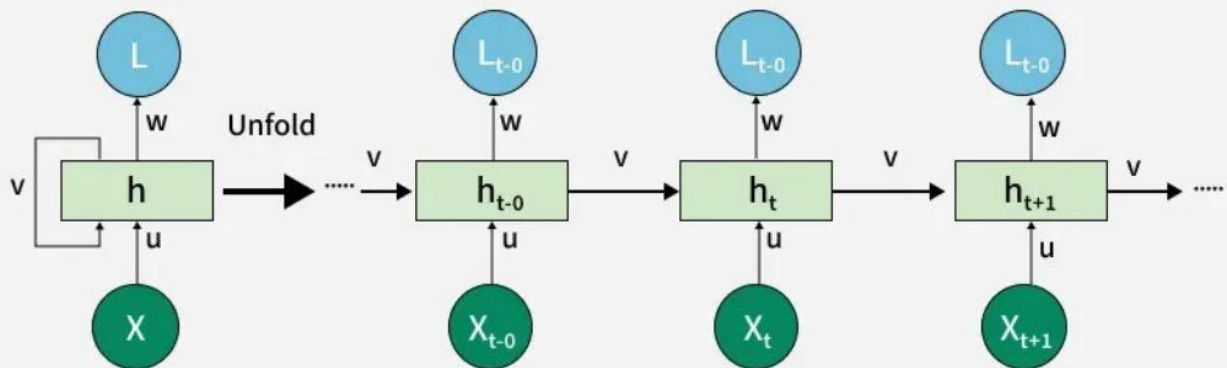


Introduction to Recurrent Neural Networks

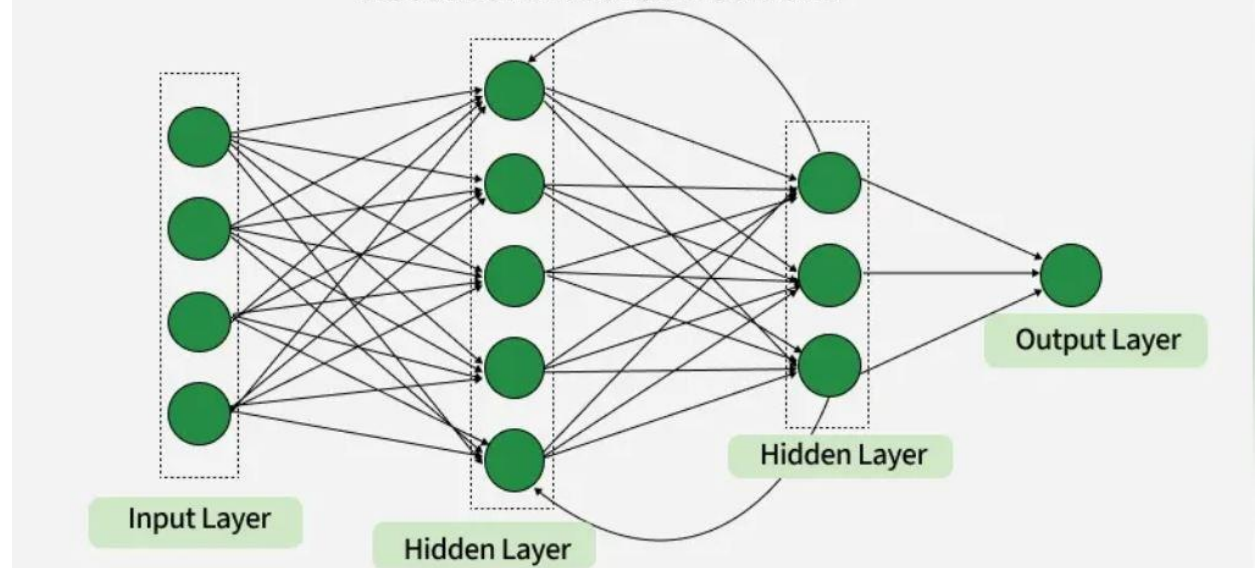
Recurrent Neural Networks (RNNs) are a class of neural networks designed to process sequential data by retaining information from previous steps. They are especially effective for tasks where context and order matter.

- Designed for sequential and temporal data
- Maintains memory of past inputs
- Widely used in NLP, forecasting and speech tasks

Introduction to Recurrent Neural Network



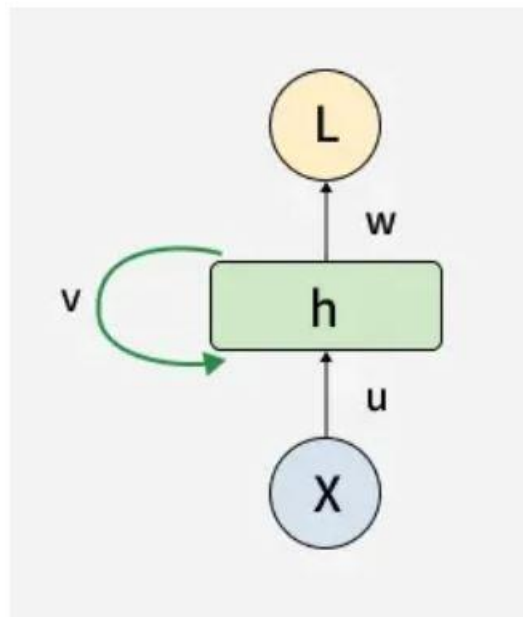
Recurrent Neural Network



Components of RNNs

1. Recurrent Neurons

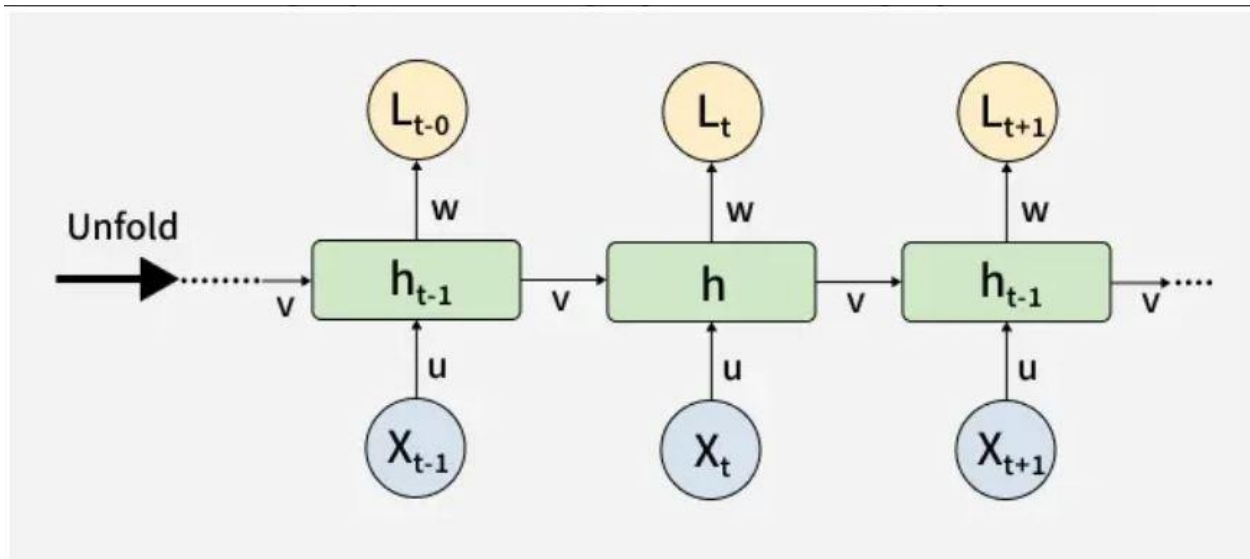
The fundamental processing unit in RNN is a Recurrent Unit. They hold a hidden state that maintains information about previous inputs in a sequence. Recurrent units can "remember" information from prior steps by feeding back their hidden state, allowing them to capture dependencies across time.



2. RNN Unfolding

RNN unfolding or unrolling is the process of expanding the recurrent structure over time steps. During unfolding each step of the sequence is represented as a separate layer in a series illustrating how information flows across each time step.

This unrolling enables [backpropagation through time \(BPTT\)](#) a learning process where errors are propagated across time steps to adjust the network's weights enhancing the RNN's ability to learn dependencies within sequential data.

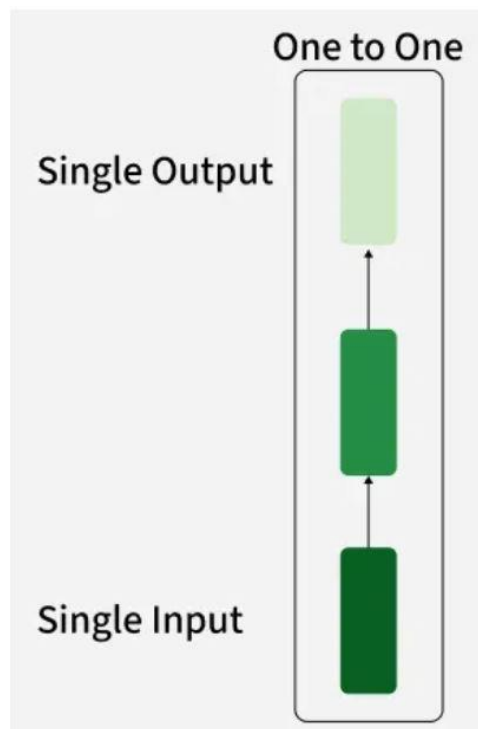


Types Of Recurrent Neural Networks

There are four types of RNNs based on the number of inputs and outputs in the network:

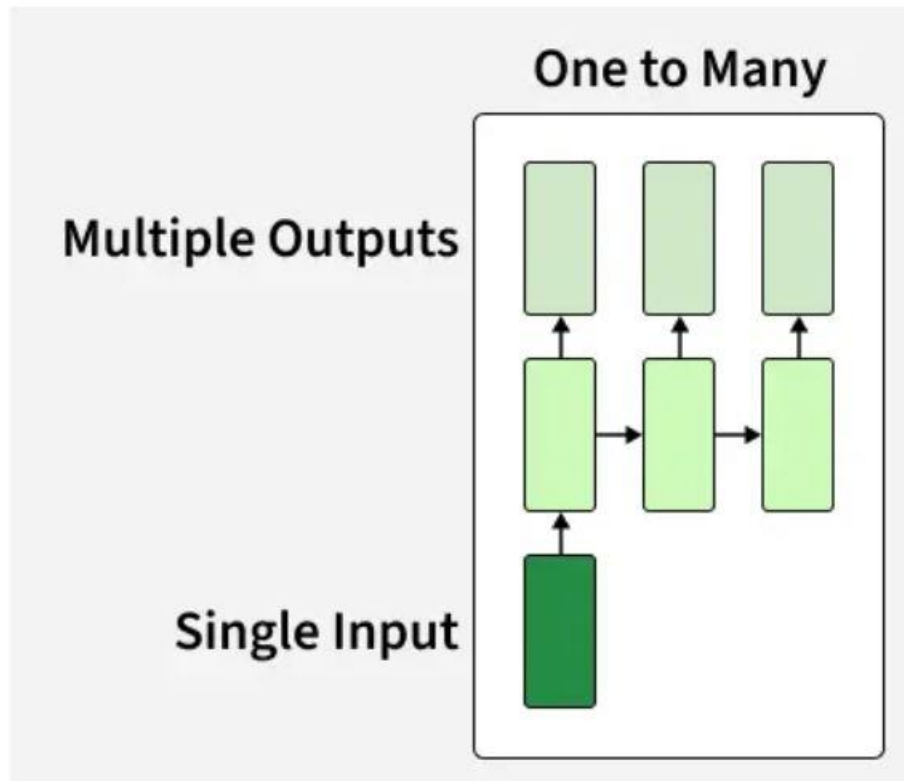
1. One-to-One RNN

This is the simplest type of neural network architecture where there is a single input and a single output. It is used for straightforward classification tasks such as binary classification where no sequential data is involved.



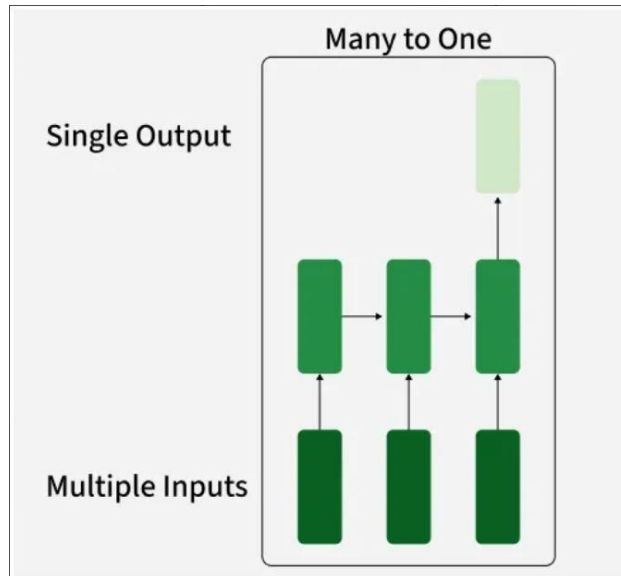
2. One-to-Many RNN

In a One-to-Many RNN the network processes a single input to produce multiple outputs over time. This is useful in tasks where one input triggers a sequence of predictions (outputs). For example in image captioning a single image can be used as input to generate a sequence of words as a caption.



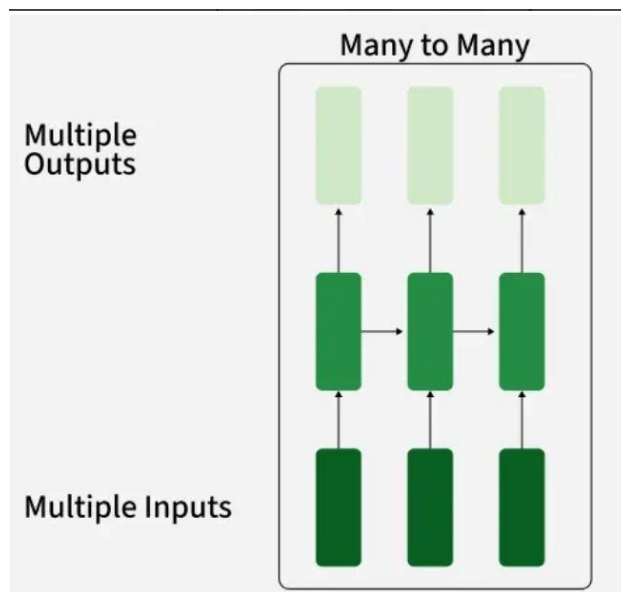
3. Many-to-One RNN

The Many-to-One RNN receives a sequence of inputs and generates a single output. This type is useful when the overall context of the input sequence is needed to make one prediction. In sentiment analysis the model receives a sequence of words (like a sentence) and produces a single output like positive, negative or neutral.



4. Many-to-Many RNN

The Many-to-Many RNN type processes a sequence of inputs and generates a sequence of outputs. In language translation task a sequence of words in one language is given as input and a corresponding sequence in another language is generated as output.



How RNN Differs from Feedforward Neural Networks?

Feedforward Neural Networks (FNNs)

- Process data in a single direction, from input to output
- Do not store information from previous inputs
- Suitable for tasks with independent data, such as image classification
- Perform poorly on sequential data due to the absence of memory

Recurrent Neural Networks (RNNs)

- Include feedback loops that pass information from previous steps
- Maintain memory of past inputs through hidden states
- Designed for sequential and time-dependent data
- Effective for tasks where context matters, such as text and time-series analysis

